

ANIRUDH KHADE

Bhopal, Madhya Pradesh

 [Portfolio](#)

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EDUCATION

National Institute of Technology

08 2024 – 08 2026

M.tech - Automation and Robotics - CGPA - 7.1

Bhopal, India

Samrat Ashok Technological Institute

09 2020 – 05 2024

B.tech - Computer Science - Internet of Things(IOT) - CGPA - 7.0

Vidisha, India

COURSEWORK / SKILLS

- | | | | |
|--------------------|---------------------------|---------------------------|----------------------|
| • Web Development | • Control system | • Python | • Gen AI |
| • Robotanalyzer | • Lean manufacturing | • Artificial Intelligence | • Internet Of Things |
| • Machine Learning | • Fundamental of robotics | • CAD/CAM | • Soft Robotics |

PROJECTs

EOG Blink Authentication via Anomaly Detection  | Presented the paper in IC2E3 05 2026

- About project Designed a passive biometric authentication system that eliminates the need for passwords on wearable devices by tracking natural eye movements (EOG) and using AI to detect anomalies..
- [Download](#)

Multimodal Parkinson's Disease Detection  | Presented the paper in NPDSM conference 26 2026

- About project This project uses AI to detect early Parkinson's disease by combining arm swing kinematics, tremor characteristics, and speech biomarkers, enabling accurate, non-invasive, and cost-effective screening..
- [Download](#)

Predictive LPG Gas Leakage Detection  | Presented the paper in NPDSM conference 26 2026

- About Project Developed a purely software-based AI framework for predictive LPG gas leakage detection and risk assessment. It integrates a soft-voting ensemble classifier with a BiLSTM network using Bahdanau attention to achieve high accuracy without relying on specialized hardware.
- [Download](#)

Automated Menstrual Cycle Anomaly Detection  | Presented the paper in NPDSM 26 2026

- About project This project detects abnormal menstrual cycle patterns by comparing 1D-CNN and Random Forest models on time-series health data. The study evaluates their effectiveness in identifying irregular cycles, supporting early health monitoring and personalized healthcare..
- [Download](#)

GestureMirror: Real-Time Robot Control via Gesture and Speech  | Presented (NPDSM) 26 2026

- About project GestureMirror is a lightweight ROS2-based framework that enables real-time human-robot interaction through gesture recognition and speech commands. It integrates multimodal inputs for intuitive, responsive, and efficient robot control in dynamic environments..
- [Download](#)

Data Quality Outweighs Model Complexity in Motor Imagery BCI  | (NPDSM) 26 2026

- About project EEGNet outperforms CSP+LDA in motor imagery BCI decoding, while EEG signal quality remains the dominant factor affecting performance and transferability..
- [Download](#)

Planar Serial Manipulators Benchmark  | 02 2027

- About project- Conducted a comprehensive multi-criteria benchmark of twelve planar serial manipulators (ranging from 2R to 5R configurations). Evaluated the effects of degrees of freedom and link-length distributions on kinematic accuracy, dynamic manipulability, and closed-loop tracking performance..
- [Download](#)

- About project **Proposed a post-training quantization pipeline designed to compress Large Language Models (LLMs) to ultra-low bit rates. The project utilizes piece-wise orthogonal rotations and Hessian-guided bit allocation to mitigate quantization errors and maintain model quality.**
- [Download](#)

INTERNSHIP

Infrared OHE Monitoring System – Kota Division [🔗](#)

12 2024-06 2025

Kota, India

- I have completed my internship in the Infrared OHE Monitoring System –Kota Division project, where I am actively volunteering, supporting, and learning. I am assisting in the installation and calibration of thermal imaging cameras, conducting GPS mapping of OHE poles, and participating in field inspections to ensure system compliance. Additionally, I am supporting the team in real-time monitoring, data collection, and report generation, analyzing temperature variations and OHE height stagger measurements. Through this experience, I am continuously enhancing my understanding of thermal imaging technology, automation, and railway safety monitoring, while also strengthening my skills in data analysis, system integration, and preventive maintenance.

TECHNICAL SKILLS

Languages: Python, Matlab, Robot Analyzer, Machine learning, JavaScript, Html, CSS, Nodejs

Developer Tools: VS Code

Technologies/Frameworks: GitHub, ReactJS, Creo, NodeJS, ExpressJS, Git, Mongo

CERTIFICATIONS

- Project- Over head equipment Monitoring system
- Outlier-Generative AI
- EOG-Blink BiometricS
- Data Quality Outweighs Model Complexity in Motor Imagery BCI
- GestureMirror: Real-Time Robot Control via Gesture and Speech
- Automated Menstrual Cycle Anomaly Detection
- Predictive LPG Gas Leakage Detection
- Multimodal Parkinson's Disease Detection